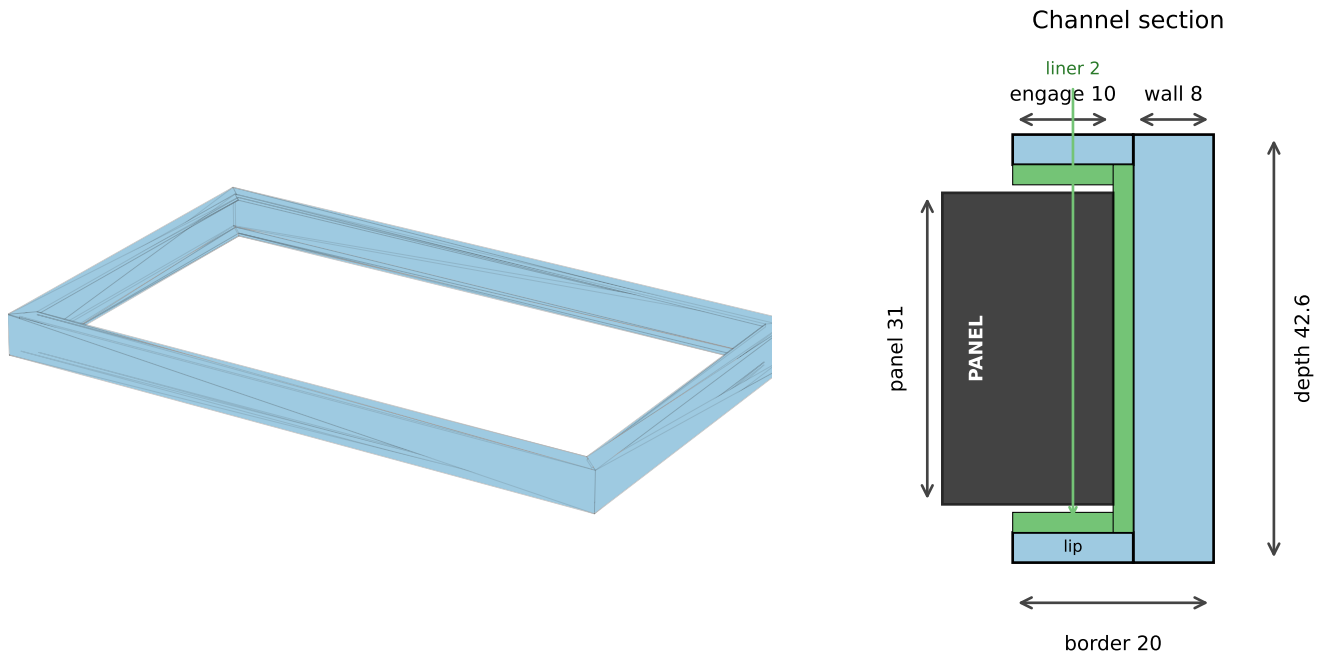


27" DISPLAY TRANSPORT FRAME

Design package — overview



Assembled frame (reference)

KEY SPECIFICATION

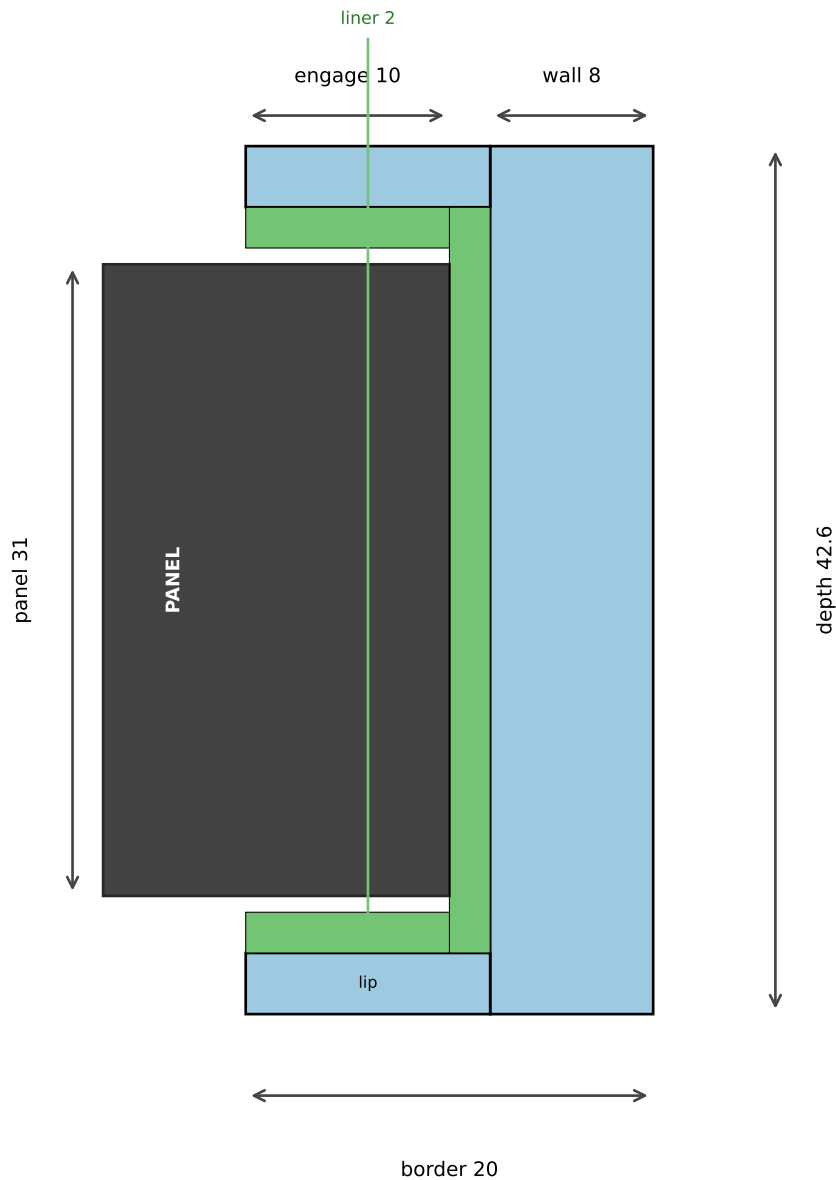
Target display	Apple Studio Display 27" (no stand / VESA)
Panel W x H x D	623 x 362 x 31 mm
Frame outer	644.2 x 383.2 x 42.6 mm
Border width	20 mm (wall 8 + engage 10 + liner 2)
Compliant liner	2 mm foam/flock/TPU bonded in groove
Visible aperture	604.2 x 343.2 mm
Fit clearance	0.6 mm in-plane / 0.8 mm per face + liner
Print bed assumed	220 mm (210 usable)
Pieces	4 corners + 6 top/bottom + 2 side = 12 total
Material (recommended)	PETG / ASA, 4 walls, 30-40% infill
Frame volume	829 cm ³ (~1028 g PLA / ~1053 g PETG)

C-channel 'picture frame': front + back lips clamp the bezel faces; the outer wall takes edge impacts. The groove is lined with compliant foam/flock for a glove-snug, non-scratch grip. The ring is split into bed-sized pieces joined by Ø3 pins + glue.

FIRST: print corner_TR only and test-fit on a real screen corner before printing the full set.

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Sheet 1 — Channel cross-section



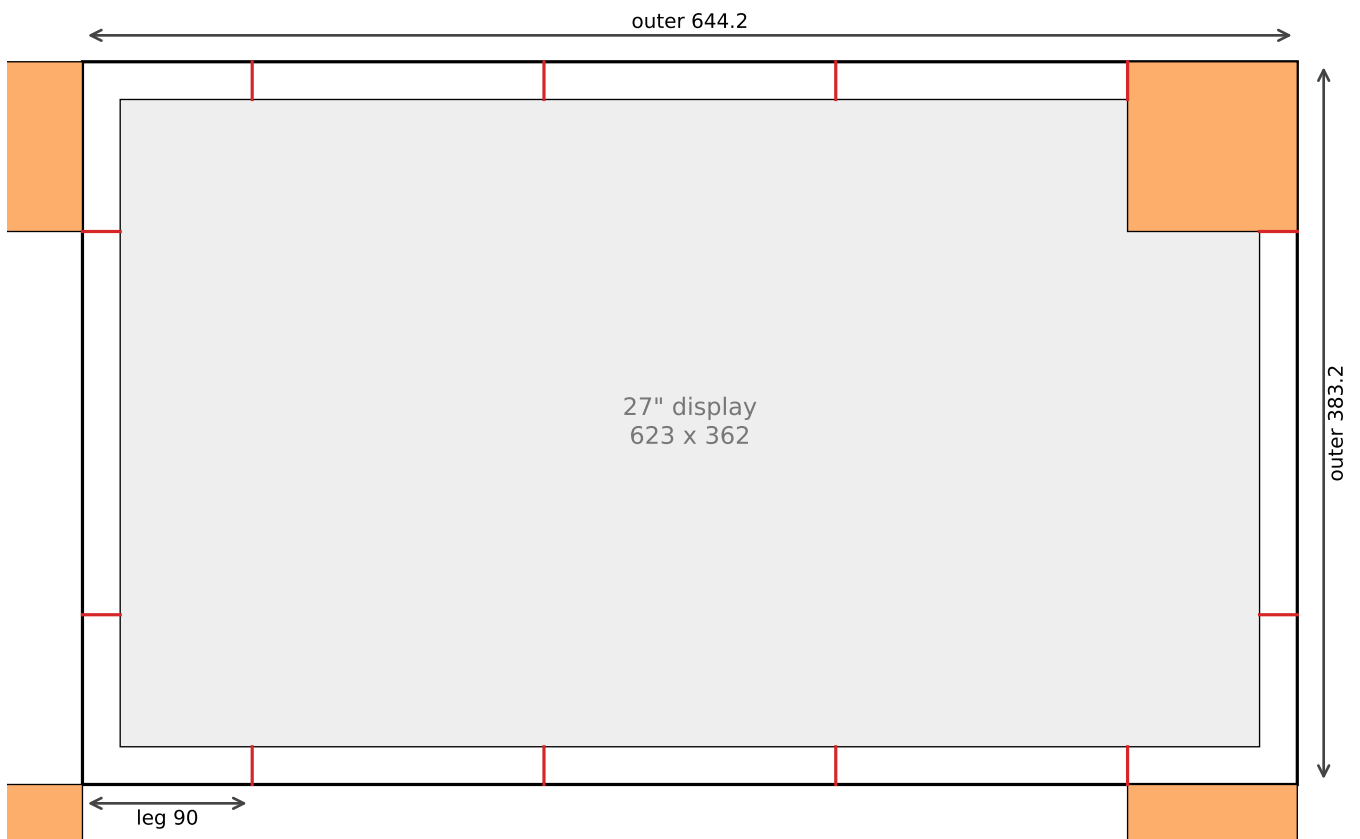
Section is constant around the entire perimeter (mitred at corners).

Tuning:

- liner_thk — foam/flock liner: the glove-snug, non-scratch grip
- clr / clr_z — hard clearance under the liner (raise if it won't seat)
- engage — how far lips wrap the bezel (also groove depth)
- wall — outer impact wall thickness
- lip_z — front/back lip thickness

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Sheet 2 — Frame plan & segmentation

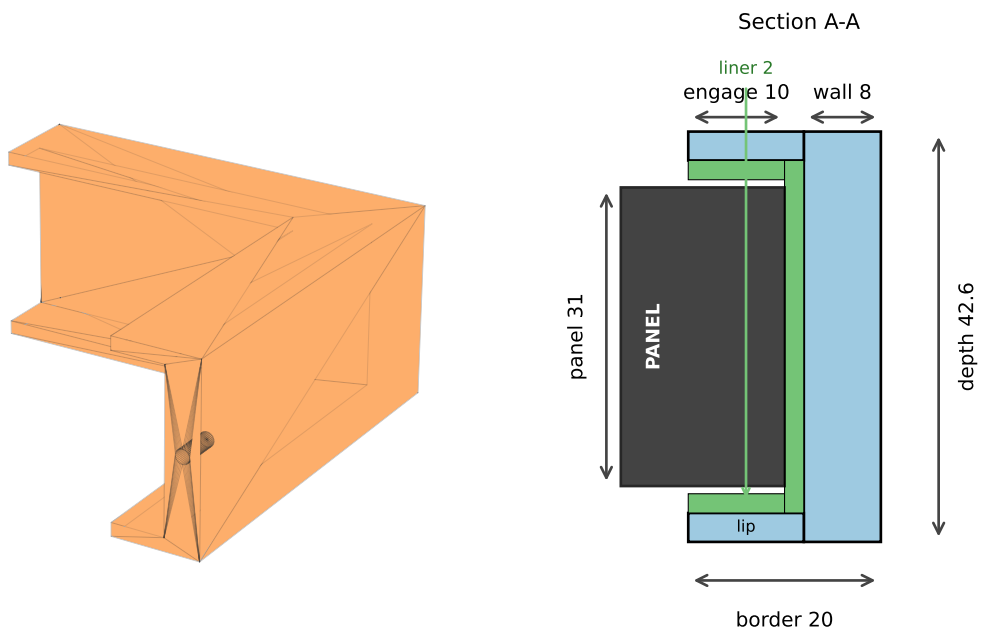
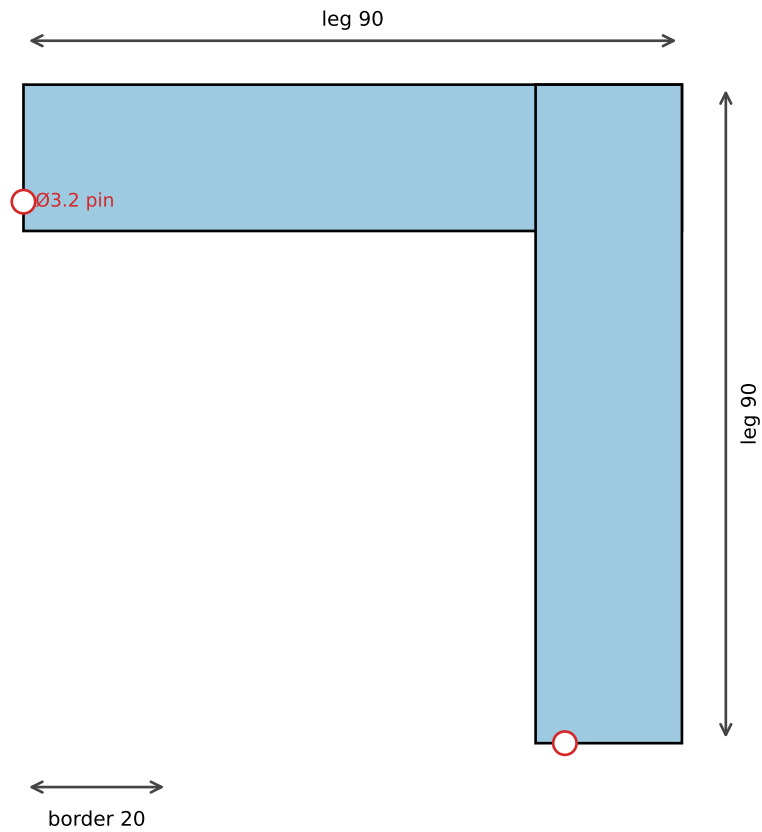


Orange = 4 corner pieces · Red = cut planes · 3 segment(s) per top/bottom run, 1 per side run

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Sheet 3 — Corner piece (x4: TR, TL, BR, BL)

Front view (looking along screen axis)

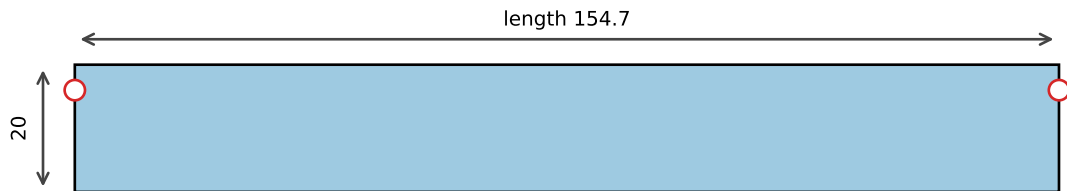


Footprint 90 x 90 x 42.6 mm · 2 pin holes per piece · print flat, no supports.

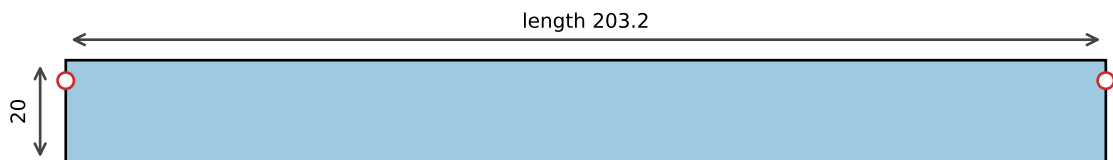
27" DISPLAY TRANSPORT FRAME

Sheet 4 — Edge segments

Top / bottom segment (x6) — pin hole each end



Left / right segment (x2) — shown straightened



All edge segments are border 20 mm x depth 42.6 mm in section (same channel as the corners). Lay flat, depth standing up.

27" DISPLAY TRANSPORT FRAME

Sheet 5 — Bill of materials & cut list

Qty	Part	STL file	Size (mm)
4	Corner (L)	corner_TR/TL/BR/BL.stl	90 x 90 x 42.6
3	Top segment	edge_top_1..3.stl	155 x 20 x 42.6
3	Bottom segment	edge_bot_1..3.stl	155 x 20 x 42.6
1	Right segment	edge_right_1..1.stl	203 x 20 x 42.6
1	Left segment	edge_left_1..1.stl	203 x 20 x 42.6
~12	Alignment pin	pin.stl (or Ø3 rod)	Ø3.0 x 17
~2.0 m	Foam/flock liner	self-adhesive tape	2 mm thick, ~10 mm wide
1	Whole-ring preview	frame_full_REFERENCE.stl	DO NOT print whole

Total printed body pieces: 12 Joints: 12 Est. filament: ~1028 g PLA / ~1053 g PETG

CONSUMABLES / HARDWARE

- ~1.1 kg PETG or ASA filament (structural; PLA OK for test fits only)
- Ø3 mm pins: printed pin.stl, or cut 3 mm rod / filament to 17 mm
- Epoxy or CA adhesive for the joints (frame is a permanent ring)
- ~2.0 m self-adhesive 2 mm EVA foam / flock tape -> the glove-snug liner in the groove
- Closed-cell foam strip to bond the outer wall to the case shell
- Optional: M3/M4 L-brackets to bolt the frame to the case ribs

27" DISPLAY TRANSPORT FRAME

Sheet 6 — Print settings & assembly

PRINT SETTINGS

- Material: PETG / ASA / ABS (impact + heat). PLA for test fits only.
- Orientation: lay flat, 42.6 mm depth standing up -> pin holes horizontal, no supports.
- Perimeters: 4 · Top/bottom layers: 5 · Infill: 30-40% gyroid.
- Layer height: 0.2-0.28 mm · nozzle 0.4-0.6 mm.
- Print one corner first and test-fit before committing the full set.

ASSEMBLY

1. Line the groove of every piece with the self-adhesive foam/flock tape (both lips + back wall) BEFORE joining.
2. Dry-fit all pieces around the display, pins in the Ø3.2 holes; the lined groove should grip with no rattle.
3. Check the ring closes square and grips evenly all round.
4. Bond each joint with epoxy/CA; pins keep the faces aligned.
5. Seat the frame in the shell; bond foam to the 8 mm outer wall, or bolt L-tabs through it into the ribs.

DIAL THE FIT (glove-snug)

- Liner too tight to seat: drop liner_thk 0.5 mm, or use softer foam.
- Still rattles: raise liner_thk, or add a second foam layer locally.
- Re-print one corner each time and test on a real screen corner.

MEASURE BEFORE FINAL PRINT

- Exact panel edge thickness (bezels taper) -> set panel_d.
- Case internal depth must clear frame depth 42.6 mm.
- Case internal W x H must clear frame outer 644 x 383 mm.

RETUNE (frame.scad / generate_frame.py)

- Different screen: panel_w / panel_h / panel_d.
- Grip: clr, clr_z. Protection: wall, engage. Printer: bed.
- Re-run: python3 generate_frame.py (STLs) and python3 design_pdf.py (this document).